Bi-Weekly Status Report 6

Project Name	CodeTrail – Real-Time Collaborative Code Editor with Contribution Tracking
Course	WS & SOA (IT645) | Dhirubhai Ambani University (Formerly DA-IICT)
Report Period	17/10/2026 – 01/11/2026 (Weeks 13–14)
Date Submitted	01/11/2026
Prepared By	Shaikh Maryam on behalf of CodeTrail Team

Project Status Summary

Sprint / Milestone	SCRUM Sprint 6 (Bi-Weekly 6) – Module 6: Live Contribution Telemetry and Module 7: Contribution Reports & PDF Export
% Complete (overall)	87.5% (7 of 8 functional modules delivered; final integration and hardening remain)

Indicator	Current	Previous	Trend	Status
Overall Project	87.5%	75%	↑	On Track
Schedule	On Schedule	On Schedule	→	On Track
Scope	7/8 Modules	6/8 Modules	↑	On Track
Quality	Analytics, report, PDF, and regression suites added; runner blocked by conflict markers	Module 5 and 6 suites passed	→	Needs Attention
Team Contribution	6 contributors with reporting-period commits	4 contributors	↑	On Track
Dependencies	Backend API, MongoDB, Socket.IO, PDF libraries and frontend environment	Backend API and analytics environment	→	Needs Attention

Executive Summary

This period completed live contribution telemetry and analytics, then added the contribution-reporting and PDF-export workflow. The team implemented authenticated report endpoints, content validation, signed Markdown/HTML report generation, client-side PDF generation, PDF structural validation, a report view, and regression coverage for workspace and execution workflows. The latest changes also corrected runtime behavior around workspace member roles, member removal, execution access, and activity-log role values. Backend tests now cover analytics, contribution reports, PDF export, runtime fixes, and workspace regressions; however,

the aggregate runner currently requires merge-conflict cleanup in `backend/tests/index.js` before final validation can pass.

Commit and Date Analysis

Table 1: Reporting-period repository evidence and module mapping

Date	Commit	Author	Module	Analysis
21 Oct.	`891581b5`, `e700f75b`	Manushri Thakkar	Integration / configuration	Added the system documentation, environment template, authentication and route corrections, and backend route mounting needed for a consistent application setup.
21 Oct.	`2683a6f1`, `71dd5d0f`	Manushri Thakkar	Frontend integration	Updated frontend dependency lock data and corrected authentication-page issues affecting login, password recovery, and reset flows.
21 Oct.	`56a7614d`, `d2bcd674`, `979afec8`	Manushri Thakkar	Deployment configuration	Centralized the frontend API base URL and made the Google OAuth client ID environment-configurable, followed by an API configuration correction.
30 Oct.	`40a75a84`	Maryam Shaikh	Module 6: Live telemetry	Added workspace analytics controllers and routes, session socket integration, telemetry constants/hooks, analytics dashboard, participation data, and cross-module workspace integration.
01 Nov.	`f3ef5ed3`	Rishika Shah	Module 6: Session analytics	Added session analytics calculations and validation, expanded analytics routes and dashboard behavior, and introduced the CT-198/CT-201 session analytics test suite.
01 Nov.	`55f7c3a0`, `22887fd8`	Anistina Dsouza	CT-199: Contribution reports	Added authenticated report-generation routes, report metrics and content validation, signed Markdown/HTML output, report tests, and the frontend contribution-report layout and validation certificate.
01 Nov.	`94b8617a`, `705ba726`	Meet Sheth	CT-200: PDF export and runtime hardening	Added client-side PDF generation and structural validation, PDF export tests, workspace and execution runtime fixes, activity-log model corrections, and regression suites; merged the feature branch into main.

Key Accomplishments in the Reporting Period

Delivered Features

- **Live Contribution Telemetry:** Connected activity, workspace, and workspace-session flows so contribution data can be collected and synchronized while a collaborative session is active. (Commit `40a75a84`)
- **Contribution Analytics APIs:** Implemented authenticated endpoints for contribution scores, coding activity, participation metrics, aggregate validation, session summaries, and standalone session validation. (Commits `40a75a84`, `f3ef5ed3`)
- **Analytics Dashboard:** Added a dedicated `ContributionAnalytics` view with contributor leaderboard, action-type breakdown, coding metrics, participation time, session information, refresh behavior, and validation status. (Commits `40a75a84`, `f3ef5ed3`)

- **Session Analytics Validation:** Added checks for duration and timestamp boundaries, participant counts and activity shares, event-category totals, duplicate events, execution rates, and mathematical consistency of session summaries. (Commit `f3ef5ed3`)
- **Regression and Edge-Case Testing:** Added tests for empty sessions, single and multi-participant sessions, zero-activity members, invalid timestamps, duplicate events, large activity datasets, category consistency, and successful or failed master validation scenarios. (Commit `f3ef5ed3`)
- **Contribution Report Generation (CT-199):** Implemented standard and custom contribution-report endpoints, report IDs and cryptographic signatures, executive metrics, contributor rankings, session details, Markdown/HTML formatting, and report-content validation. Added a dedicated contribution-report view with validation-certificate presentation. (Commits `55f7c3a0`, `22887fd8`)
- **PDF Export and Validation (CT-200):** Added branded client-side PDF generation with multi-page tables, metadata, page numbering, report signatures, safe filenames, and structural/content validation for generated PDFs. (Commits `94b8617a`, `705ba726`)
- **Runtime and Regression Hardening:** Corrected workspace member-role and removal handling, canonicalized activity-log roles, secured execution access checks, and added PDF-export, runtime-fix, and workspace-regression test suites. (Commits `94b8617a`, `705ba726`)
- **Application Configuration and Documentation:** Improved route mounting, frontend API configuration, OAuth environment handling, authentication-page fixes, dependency lock data, and system documentation. (21 October commits)

Planned vs. Actual

Task/ Deliverable	Planned Completion	Current Forecast	Actual Status	Notes
Module 6: Live Telemetry	Oct. 26	—	Done	Workspace activity synchronized through backend and Socket.IO flows
Contribution Analytics APIs	Oct. 30	—	Done	Contribution, coding, participation, and validation routes implemented
Session Analytics	Nov. 1	—	Done	Session summaries and participant-level metrics added
Analytics Validation Suite	Nov. 1	—	Done	CT-198/CT-201 edge cases and consistency checks added
Contribution Report Generation	Nov. 1	—	Done	CT-199 report APIs, signed content, validation, and report view added
PDF Export and Validation	Nov. 1	—	Done	CT-200 PDF generation, structural checks, and export tests added
Final integration and demonstration	Nov. 8	Nov. 8	In Progress	Remaining work is release hardening, documentation, and demonstration preparation

Project Risks, Issues, & Dependencies

Type (ID)	Description	Impact	Owner	Mitigation / Next Step
Dependency (D-04)	Reports and PDF exports require authenticated backend access, workspace data, configured session data, and frontend PDF dependencies	High	Meet Sheth / Maryam Shaikh	Validate the deployed environment and repeat end-to-end checks for JSON, Markdown, HTML, and PDF formats.
Issue (I-06)	Analytics responses combine live session data and aggregated activity data, creating consistency risk during concurrent updates	Medium	Rishika Shah	Keep server-side validation checks and compare aggregate values with raw activity events.
Risk (R-07)	Large event histories can increase analytics response and rendering time	Medium	Frontend / Backend team	Preserve bounded queries and event streams, then profile and paginate analytics surfaces if required.
Risk (R-08)	Final integration now spans execution, telemetry, audit, analytics, report generation, and PDF export modules	Medium	Project manager	Use the commit-to-module map and run the complete regression suite before demonstration.
Issue (I-07)	The backend test index contains unresolved Git conflict markers after feature integration	High	Meet Sheth / Project manager	Resolve the conflict markers, preserve all intended test imports, and rerun the complete backend suite.

Project Decisions

Project Scope Changes

Change	Action	Reason	Schedule Impact	Approved By
Module 6 analytics and live telemetry	Added	Implements the approved functional module for contribution and session reporting.	None	Team
Contribution reports and PDF export	Added	Extends analytics into signed, validated, downloadable contribution-report artifacts (CT-199/CT-200).	None	Team

Upcoming Work (Next Two Weeks' Plan – Bi-Weekly 7 Target)

- Perform end-to-end validation of live workspace sessions and contribution reports with the backend, database, Socket.IO, and frontend running together.
- Validate JSON, Markdown, HTML, and PDF report exports using populated, empty, and large workspaces.
- Profile analytics queries, report generation, PDF rendering, and dashboard behavior for large activity histories.
- Resolve remaining frontend lint or integration warnings and review production bundle behavior.

Decision	Rationale	Impact	Decided By	Date
Expose analytics through authenticated workspace routes	Analytics contains participant and contribution information and must follow the existing workspace authorization boundary.	Consistent access control across analytics endpoints.	Team	Oct. 30
Use a dedicated analytics dashboard	A separate view keeps contribution, coding, participation, and session metrics discoverable without overloading the workspace editor.	Clearer reporting and easier demonstration of Module 6.	Team	Oct. 30
Validate aggregates against raw activity data	Summary percentages, line counts, event categories, and execution rates require mathematical consistency checks.	More trustworthy contribution reporting and diagnosable failures.	Rishika	Nov. 1
Support empty and partial sessions explicitly	New sessions may have no participants, events, or executions and must not fail with divide-by-zero or invalid timestamp errors.	Safe initial-state behavior and stronger edge-case coverage.	Team	Nov. 1
Sign contribution reports cryptographically	Report signatures provide an integrity check for generated contribution summaries and exported artifacts.	Report tampering can be detected during validation and audit review.	Anistina	Nov. 1
Validate PDF structure before export completion	Generated files must contain valid PDF markers, page structure, required report content, and preserved signatures.	More reliable downloadable reports and clearer diagnostics for corrupted output.	Meet	Nov. 1

- Complete final integration testing across execution, telemetry, audit, analytics, report, and PDF workflows.
- Finalize technical documentation, user demonstration material, and submission evidence.

Team Member Contributions

Each member performed manual unit testing for their assigned components. The designated members also created test cases and contributed to the cumulative test documentation.

Manushri Thakkar — Integration and Configuration

Name: Manushri Thakkar (202512122)

Role: Project Manager / DevOps / Backend Developer

Tasks Completed	Corrected backend route mounting and authentication-related integration issues, added the frontend API configuration and environment example, enabled configurable Google OAuth, consolidated system documentation, and created test cases and cumulative test documentation for the completed work.
Contribution Type	Development / Integration / Documentation
Hours / Capacity	30/30 hrs logged
Evidence	`backend/index.js`, `frontend/src/config/api.js`, `frontend/.env.example`, `Documents/CodeTrail_System_Documentation.md`; commits `891581b5`, `e700f75b`, `56a7614d`, `d2bcd674`, `979afec8`
Testing	Performed manual unit testing of the configuration, authentication, and route changes.

Shaikh Maryam — Telemetry Integration

Name: Shaikh Maryam (202512074)

Role: Full-Stack Developer / UI Architect

Tasks Completed	Led the Module 6 integration by connecting activity and workspace-session telemetry, implementing analytics controller and route integration, updating workspace and dashboard surfaces, and coordinating the live contribution telemetry delivery. Also debugged and resolved major Module 5 telemetry, audit, workspace-integration, and execution-interface issues during integration.
Contribution Type	Development / UI Architecture / Integration / Debugging
Hours / Capacity	30/30 hrs logged
Evidence	`backend/controllers/analyticsController.js`, `backend/sockets/workspaceSessionSocket.js`, `frontend/src/hooks/useContributionTelemetry.js`, `frontend/src/modules/analytics/ContributionAnalytics.jsx`, `frontend/src/modules/telemetry/ContributionDossier.jsx`; commits `40a75a84`, `4cfc152b`, `47b4e834`, `55321f9e`
Testing	Combined backend Module 5 and Module 6 test suite completed successfully; performed manual unit testing of telemetry synchronization, analytics routes, dashboard integration, and the Module 5 fixes.

Mahi Kansara — Socket.IO and Analytics Support

Name: Mahi Kansara (202512078)

Role: Frontend Developer / Socket.IO Integration Lead

Tasks Completed	Supported the live telemetry and Socket.IO integration, refined the analytics dashboard and contribution data flow, and created test cases and contributed to the cumulative test documentation.
Contribution Type	Development / Testing / Documentation
Hours / Capacity	30/30 hrs logged
Evidence	`backend/utils/telemetryStore.js`, `frontend/src/hooks/useContributionTelemetry.js`, `frontend/src/modules/analytics/ContributionAnalytics.jsx`
Testing	Performed manual unit testing of Socket.IO synchronization, telemetry updates, and analytics dashboard behavior; created test cases and contributed to the cumulative test documentation.

Rishika Shah — Session Analytics

Name: Rishika Shah (202512041)

Role: Frontend Developer / Analytics and Validation Lead

Tasks Completed	Implemented session analytics and participant activity validation, expanded the analytics dashboard with session information and validation behavior, and created the comprehensive CT-198/CT-201 test suite.
Contribution Type	Development / Testing / Analytics
Hours / Capacity	30/30 hrs logged
Evidence	`backend/tests/sessionAnalytics.test.js`, `backend/utils/analyticsValidator.js`, `frontend/src/modules/analytics/ContributionAnalytics.jsx`; commit `f3ef5ed3`
Testing	Session analytics tests cover empty, single-participant, multi-participant, timestamp, deduplication, category, execution, and large-dataset cases; performed manual unit testing of session analytics, validation feedback, and dashboard behavior.

Meet Sheth — PDF Export and Runtime Hardening

Name: Meet Sheth (202512057)

Role: Backend Architect / System Designer

Tasks Completed	Supported the backend architecture and workspace integration required for the new authenticated analytics and live session telemetry routes.
Contribution Type	Architecture / Integration
Hours / Capacity	30/30 hrs logged
Evidence	`backend/routes/workspaceRoutes.js`, `backend/routes/workspaceSessionRoutes.js`, `backend/controllers/workspaceSessionController.js`, `frontend/src/utils/pdfReportGenerator.js`; commits `94b8617a`, `705ba726`
Testing	Implemented PDF export and structural-validation functionality, added PDF export and workspace regression tests, and performed manual unit testing of the workspace-session integration, execution access, and PDF report behavior.

Dudhat Monar — Quality Assurance

Name: Dudhat Monar (202512019)

Role: Backend Developer / QA Engineer

Tasks Completed	Continued the integrity and validation focus from Module 5 while the analytics pipeline was extended to include cross-checks against activity data. Also created test cases and contributed to the cumulative test documentation.
Contribution Type	Testing / Quality Assurance
Hours / Capacity	30/30 hrs logged
Evidence	`backend/tests/integrityAuditor.test.js`, `backend/utils/cryptoProof.js`
Testing	Performed manual unit testing of integrity, validation, and analytics behavior; created test cases and contributed to the cumulative test documentation.

Anistina Dsouza — Contribution Reports

Name: Anistina Dsouza (202512081)

Role: Full-Stack Developer / Module Lead

Tasks Completed	Implemented CT-199 contribution-report generation, report-content validation, signed Markdown/HTML output, the contribution-report view, and validation-certificate presentation, while supporting regression coverage for the existing execution and telemetry modules.
Contribution Type	Development / Testing / Documentation
Hours / Capacity	30/30 hrs logged
Evidence	`backend/tests/executionValidation.test.js`, `backend/tests/telemetrySync.test.js`, `backend/controllers/reportController.js`, `backend/utils/reportGenerator.js`; commits `55f7c3a0`, `22887fd8`
Testing	Implemented CT-199 report generation and content-validation tests, added the contribution-report view, and performed manual unit testing of execution, telemetry, report-generation, and regression scenarios.

Ruchita Patadiya — Dossier and Audit Export

Name: Ruchita Patadiya (202512070)

Role: Frontend Developer / Backend Developer

Tasks Completed	Supported the existing contribution dossier and audit-export surfaces that provide the activity data consumed by the new analytics and reporting workflow. Also created test cases and contributed to the cumulative test documentation.
Contribution Type	Development / Integration
Hours / Capacity	30/30 hrs logged
Evidence	`frontend/src/modules/telemetry/ContributionDossier.jsx`, `frontend/src/utils/auditExport.js`
Testing	Performed manual unit testing of contribution-dossier, telemetry, and audit-export workflows; created test cases and contributed to the cumulative test documentation.

Validation Evidence

Current Validation Position

- Backend command executed: `cd backend && npm test`.
- Result: the expanded test runner is currently blocked before execution because `backend/tests/index.js` contains unresolved Git conflict markers.
- The test runner includes proof-of-work, activity-controller, telemetry synchronization, integrity auditing, contribution analytics, session analytics, CT-199 report generation, CT-200 PDF export, runtime-fix, and workspace-regression suites.
- Manual unit testing was performed by every team member for the components and workflows assigned to them.
- Final validation action: resolve the test-index conflict, then rerun `cd backend && npm test`.